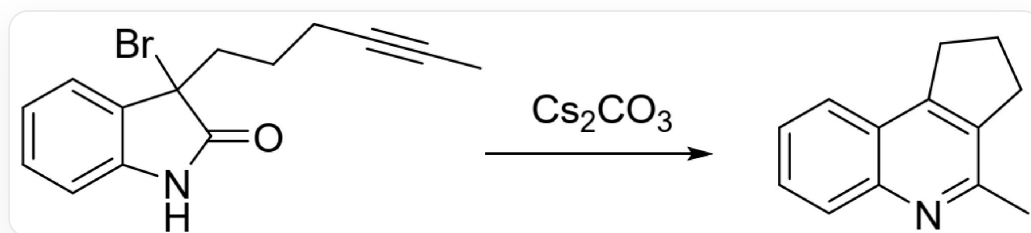


Question

Pericyclic reactions are a class of reactions with extremely high application value. Two reactions are given.

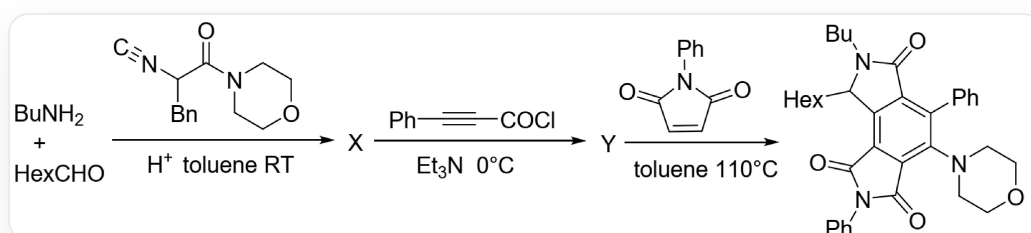
The first reaction is



CC#CCCCC1(Br)C2=CC=CC=C2NC1=O reacts in the presence of Cs2CO3 to yield CC1=NC2=CC=CC=C2C3=C1CCC3

, in which two intermediates **A** and **B** exist during this transformation.

The second reaction is



n-Butylamine, n-heptanal, and [C-]#[N+]C(CC1=CC=CC=C1)C(N2CCOCC2)=O react at room temperature in acidic toluene solvent to obtain intermediate **X**. Intermediate **X** and O=C(C#CC1=CC=CC=C1)Cl react in the presence of triethylamine at 0 degrees Celsius to obtain intermediate **Y**. Intermediate **Y** and O=C(N1C2=CC=CC=C2)C=CC1=O react in toluene solvent at 110 degrees Celsius to obtain the product CCCCCCC1C2=C3C(C(N(C4=CC=CC=C4)C3=O)=O)=C(N5CCOCC5)C(C6=CC=CC=C6)=C2C(N1CCCC)=O

Which of the following statements is correct?

A. Intermediate **A** contains 3 rings

B. The intermediate **A** has a chiral carbon.

- C.** The intermediate **B** has 13 C atoms
- D.** There is a carbon-carbon triple bond in the intermediate **B**.
- E.** There is 1 carbonyl group in the intermediate **X**
- F.** Intermediate **X** contains two N atoms with different nucleophilicities.
- G.** The intermediate **Y** contains a γ -lactam structure.
- H.** During the process of obtaining the intermediate **Y**, CO is produced.
- I.** None of the above options are correct

Answer

Correct Answer: **G**

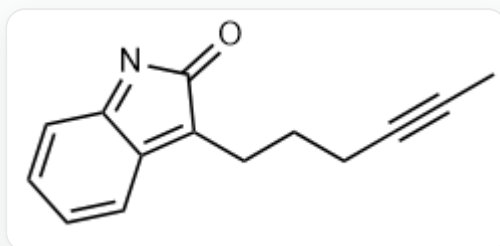
Detailed Explanation

In the first reaction, the electrons from product N are donated to the benzene ring, Br⁻ departs, and the carbonate neutralizes the H on N⁺.

CHECKPOINT

1 PTS

The diene intermediate A is obtained, with the structure


CC#CCCC1=C2C=CC=CC2=NC1=O

Thus, intermediate **A** has only two rings and no chiral carbon.

CHECKPOINT

1 PTS

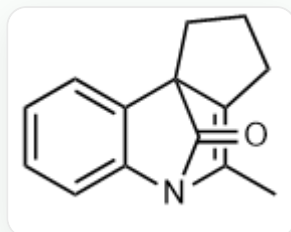
Intermediate **A** has only two rings and no chiral carbon

The alkyne undergoes a DA reaction with the conjugated diene on the five-membered ring of intermediate **A**,

CHECKPOINT

1 PTS

Intermediate **B** is obtained, with the structure


CC1=C2CCCC23C4=CC=CC=C4N1C3=O

CHECKPOINT

1 PTS

Intermediate **B** has 14 C atoms and no carbon-carbon triple bond

In the second reaction,

CHECKPOINT

1 PTS

n-Butylamine reacts with n-heptanal to form an imine by dehydration

The imine is nucleophilically attacked by the C atom of isocyanide,

CHECKPOINT

1 PTS

The imine is nucleophilically attacked by the C atom of isocyanide

Subsequently, the O atom on the carbonyl group of the isocyanide molecule attacks the carbon of the nitrilium ion, and the keto form transforms into the enol form, yielding an intramolecular cyclization product.

CHECKPOINT

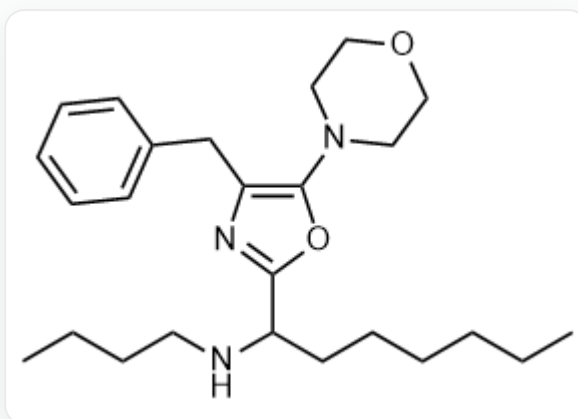
1 PTS

The O atom attacks the carbon of the nitrilium ion, undergoing an enolization transformation

CHECKPOINT

1 PTS

Intermediate **X** is obtained, with the structural formula



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CCCCCCC(C1=NC(=C(N2CCOCC2)O1)CC3=CC=CC=C3)NCCCC
```

CHECKPOINT

1 PTS

Intermediate **X** lacks a carbonyl group and has 3 N atoms with different nucleophilicities

The N atom of the butylamino group in intermediate **X** nucleophilically attacks the acyl chloride group of phenylpropynoyl chloride, resulting in the departure of Cl,

CHECKPOINT

1 PTS

The N atom of the butylamino group in intermediate **X** nucleophilically attacks the acyl chloride group of phenylpropynoyl chloride

Afterward, the $C=N-O-C=C$ moiety of the oxazole ring undergoes a DA pericyclic reaction with the alkyne, followed by a retro-DA reaction, eliminating aniline.

CHECKPOINT

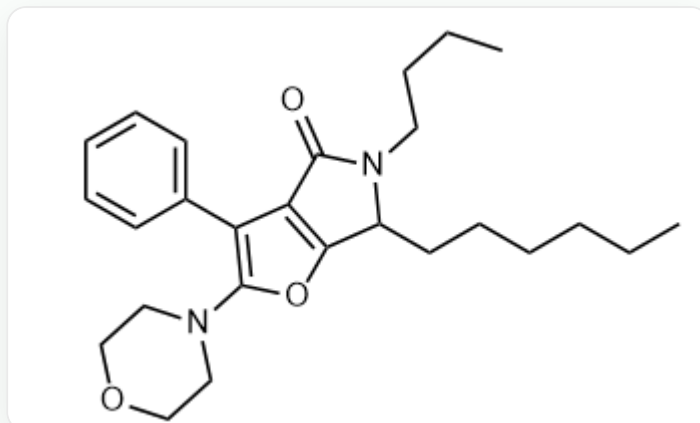
1 PTS

The oxazole ring undergoes a DA pericyclic reaction with the alkyne

CHECKPOINT

1 PTS

A retro-DA reaction occurs, eliminating aniline and yielding intermediate **Y**, with the structural formula


CCCCCCC1C2=C(C(N1CCCC)=O)C(C3=CC=CC=C3)=C(O2)N4CCOCC4

CHECKPOINT

1 PTS

Intermediate **Y** contains an amide structure, eliminating aniline rather than carbon monoxide

The final product is G.